

SP-115-Red PBI Sales

Final Report

CS 4850 Section 02

Spring 2026

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Website: <https://asparagus-goose-2rll.squarespace.com/>

GitHub: <https://github.com/KSU-Senior-Project-2026-SP-115-Red>

SP-115-Red PBI Sales

STATS AND STATUS		
Lines of Code (LOC)	65	
Components/Tools	Power BI, Excel, SquareSpace, MS Word, Power Query	
Hours Estimate	250	
Hours Actual	160	
Status	The project is 100% complete	

Table of Contents

1.0 Introduction.....	5
2.0 Challenges.....	5
3.0 Requirements	6
3.1 Functional Requirements	6
3.2 Non-Functional Requirements	8
4.0 Analysis and Design	9
5.0 Development.....	10
5.1 Development Phase.....	13
5.2 Project Setup	16
6.0 Test Plan.....	17
6.1 Overview of Test Plan	17
6.2 Scope of Test Plan	18
6.3 Progression Test Plan.....	18
6.4 Regression Test Plan.....	19
6.5 Test level responsibility	20
6.6 Test Approach.....	20
6.7 Build Strategy	21
6.8 Facility, data, and resource provision plan	21
6.9 Testing Handover Procedure.....	23

6.10 Testing Metrics	23
6.11 Test Environment Diagram.....	24
6.12 Testers	24
6.13 Hardware and Firmware	25
6.14 Software	25
6.15 Interface	25
6.16 Other Materials	26
6.17 Environment Control	26
6.18 Team Role Responsibilities	26
6.19 Assumptions.....	27
6.20 Dependencies	27
6.21 Entry and Exit Criteria.....	28
7.0 Test Report.....	28
7.1 Security	28
7.2 Stress & Volume Testing (S&V)	28
7.3 Connectivity Testing (CT)	29
7.4 Disaster Recovery/Back Up.....	29
7.4 Unit Testing	29
7.5 Integration Testing.....	29
7.6 Test Execution Schedule.....	30

7.7 Testing Tools	30
7.8 Establishing Environment.....	30
7.9 Approvals.....	31
7.10 Test Milestones and Schedule.....	31
7.11 Training.....	32
7.12 Defect Management.....	32
8.0 Version Control.....	33
9.0 Conclusion	33
Appendix.....	34

1.0 Introduction

This document will discuss the process of the Nike Sales Power BI (PBI) dashboard project. It begins with the planning phase, where it will discuss the requirements and designs. It is closely referenced to the Software Requirements Specification (SRS) and the Software Design Document (SDD). It discusses the functional and non-functional requirements that the project focuses on. It would then go into the development process and the methodologies that will be mentioned in the later sections. Then there will be testing reports after the development phase to ensure that the website is meeting the requirements and expectations from the documents. The document would go into more detail in its appropriate sections, including tables and images for easier understanding and organization.

2.0 Challenges

Challenges the team encountered throughout the project included a synthetic dataset provided by Kaggle, limited features in PBI's free version, and embedding the dashboard on the public website. The Nike Sales dataset used for the project is sourced from the Kaggle website. The dataset is synthetic, which is generated data with no collection of personal data from real customers. It still mimics real-world data patterns, and it can still provide an educational purpose for users to use and understand data analytics.

Another limitation was the limited features. While the team was brainstorming and sketching labels for the website, we ran into the problem of not meeting or finding any

similar visualizations. We would have to redo the visualizations to represent the same KPIs and ensure that users could still understand the visualizations.

The third challenge was the difficulty of embedding into the public website. The embedding process took longer than expected because it was locked behind the “Power BI admin,” or we needed to have pro access to publicly embed on the website. There was a solution that we had come up with, and we were able to successfully embed the dashboard into the website.

3.0 Requirements

3.1 Functional Requirements

Functional Requirements such as data sources, metrics and KPI, and the viewability and accessibility for users are discussed in this section. It goes into detail about each section and how the Nike Sales PBI Dashboard behaves while users are accessing the website.

3.1.1 Data Sources

The data source, Nike Sales, is sourced from Kaggle. As stated in the dataset’s description, it is synthetic and intended for educational purposes. It does not include real Nike customer data. This dataset will be cleaned first through Power Query before being imported into PBI. After importing, the data is used to visualize sales information for users to understand from an analytical perspective.

3.1.2 Metrics and KPI

The dashboard includes metrics and KPI, such as MRP, sales by product lines, gender-specific products, revenue, and discounts applied to these products. These metrics can be changed based on the user's interaction with the dashboard's features, such as slicers and filters. The dashboard allows users to see the sales pattern and trend across different categories and time periods.

3.3 User Roles and Permissions

This interactive Nike sales Dashboard consists of two user roles:

1. Viewer
2. Administrator

Viewer:

Viewers are the end-users who are using the dashboard to see the analysis of the Nike Sales dataset. They can use interactive features to cross-compare products and analyze patterns and trends. They will not have access to changing the dashboard's overlay or permission to edit the current database that is established on this dashboard.

Administrator:

Administrators consist of users, who are the developers of this dashboard. They have the same permissions to use interactive features on the dashboard to do data analysis on Nike sales, as well as the viewers. Unlike viewers, they have permission to edit the overlay of the dashboard, visuals, interactive features, and manage how data behaves to improve precision and accuracy, but will not change the source itself.

3.2 Non-Functional Requirements

The non-functional requirements section goes into two requirements that the PBI Dashboard mainly focuses on to provide users with a smooth experience. It discusses the usability for users when accessing the website and performance.

3.2.1 Usability

The PBI Nike Sales Dashboard will be accessible for first-time and returning users. Users can access this dashboard through a website.

For first-time users, the dashboard has a hovering feature. This feature allows the dashboard's KPI description to be viewable for users. It teaches users about the features provided and interacts with the dashboard.

There are interaction buttons with check boxes for users to interact with to see how data behaves with these features. If the user checks off a feature, such as slicers or filters, that interaction will be applied to the dataset and provide an updated visual representation of the current graphs. Users can check or uncheck the boxes to see the differences between the interactions to learn more about the sales performance.

There are three tabs for users to access and navigate through the dashboard. The three tabs will navigate users through different categories of Nike sales, sales representation, product lines, and combine both products. This helps users navigate through different sales representations of different products.

3.2.2 Performance

The dashboard interacts with users through the features provided, which are applied to the data graphs. Based on the user's responses to the interaction features, the dashboard responds to each interaction, updating graphs and providing an accurate representation of sales.

Before releasing to the public, the Nike sales dataset from Kaggle undergoes a data cleaning process with Power Query. Cleaning the data helps the process of visualizing data into graphs/representations with PBI and minimizes delays upon import. Data cleansing provides a cleaner and more accurate visualization of the data, ensuring users have a smooth analytical experience.

4.0 Analysis and Design

The Nike Sales PBI Dashboard follows a sequential process as seen in Figure 1. The process would begin with the developers obtaining the dataset from Kaggle. Then the downloaded dataset is imported into the Microsoft Power BI application. The dataset undergoes the data cleaning and transformation process in the application with Power Query to remove any duplicated data and to improve the consistency and accuracy before data analysis. Once the cleaning process is completed, the dataset can be used to create the dashboard with different visual representations. Interaction features are also implemented at this stage, allowing the representation to be modified upon selection to view the data behavior.

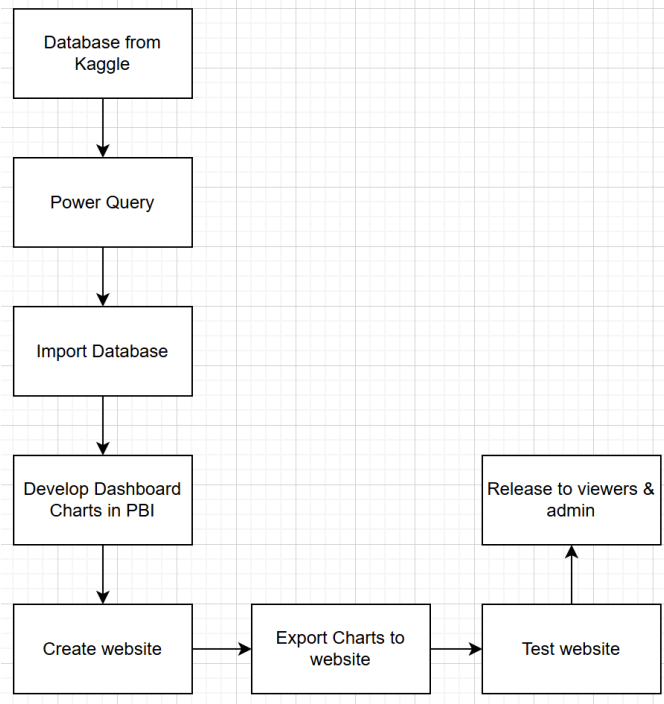


Figure 1. Architecture of Nike Sales PBI Dashboard

The Nike Sales Dashboard follows these steps to ensure that the dashboard is consistent with its appropriate categories, making the website easy to use for new and returning users. The dataset would not be affected by the interactive features or the users accessing the website. The dataset would remain constant unless the developer decides to forego further cleaning of the data. The users (or the viewers) would not have access to the Nike Sales dataset to prevent inaccuracies and unnecessary modifications that could affect the integrity of the website.

5.0 Development

The development process for the Nike Sales PBI Dashboard used the waterfall methodology. The waterfall methodology is a linear and very structured process, where we

had to complete one step completely before we move on to the next step. The process begins with the sketches and planning with the dataset, dashboard layout, functional, non-functional requirements, and more that were mentioned in the previous sections. The next step is the development process. This section is where it will go into the data handling and the development of the dashboard. It would then branch out to the interaction features that users will use on the dashboard. Then it starts into the implementation phase, where the PBI dashboard will be embedded into the website. The testing phase will be discussed in the next section, now that the development process of the dashboard is completed. The process would repeat for each tab to ensure consistency, as seen in sketches from Figures 2, 3, and 4.

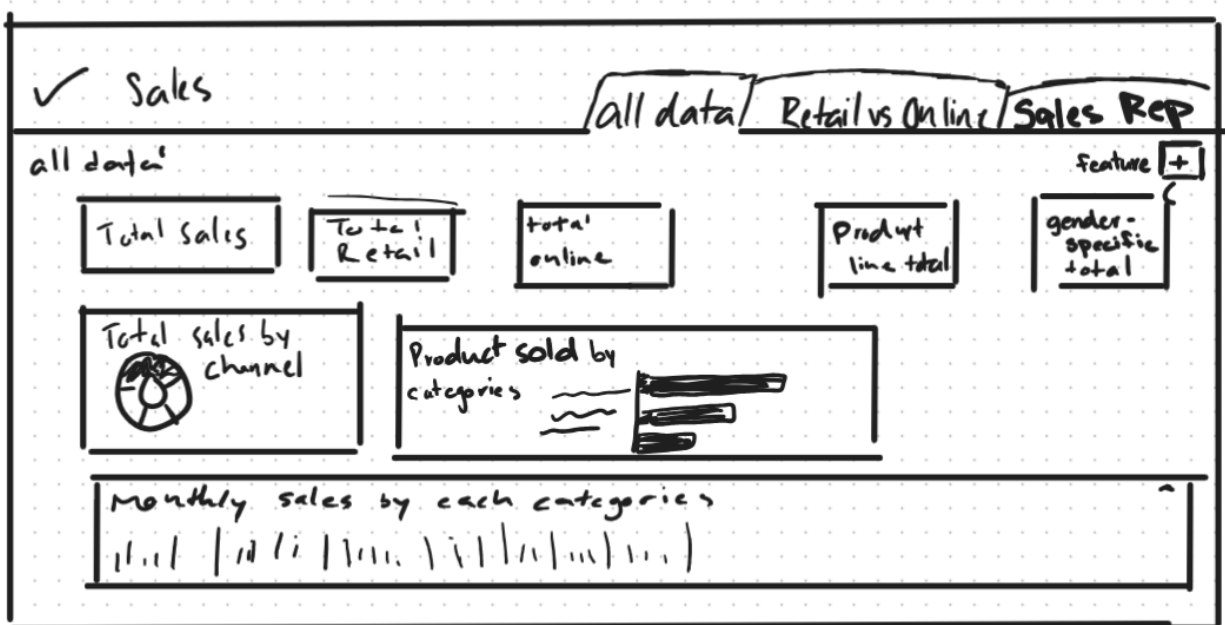


Figure 2. First Tab "All Data"

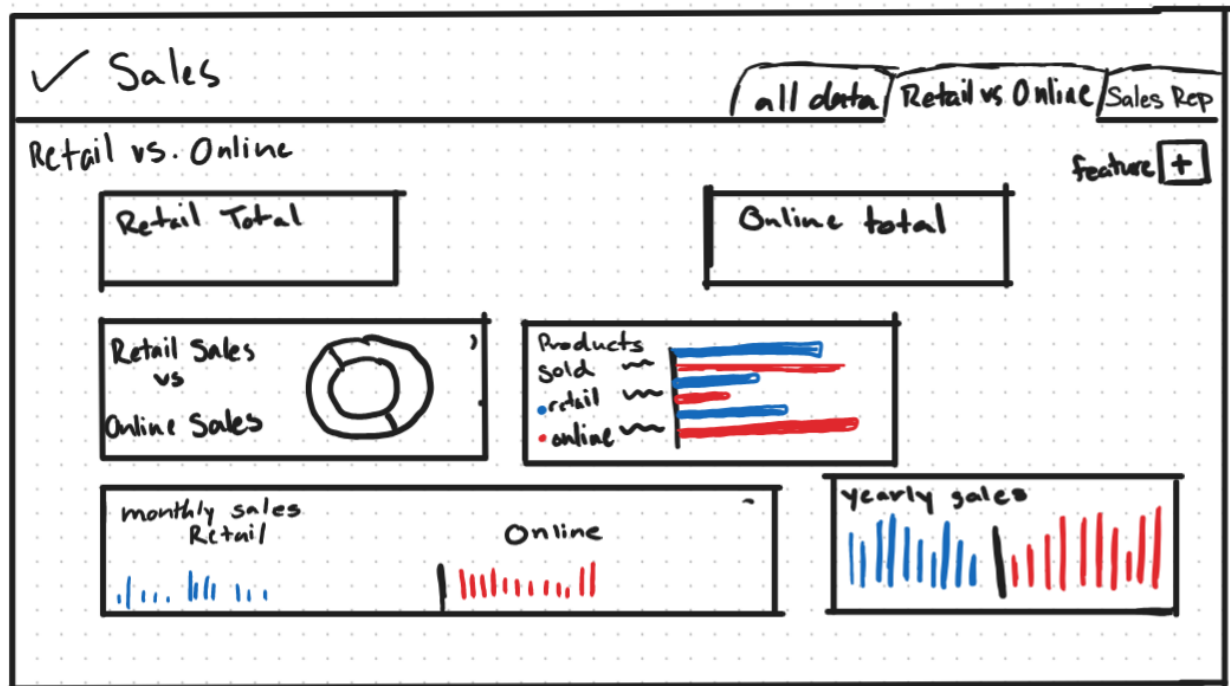


Figure 3. Second Tab “Retail vs. Online”

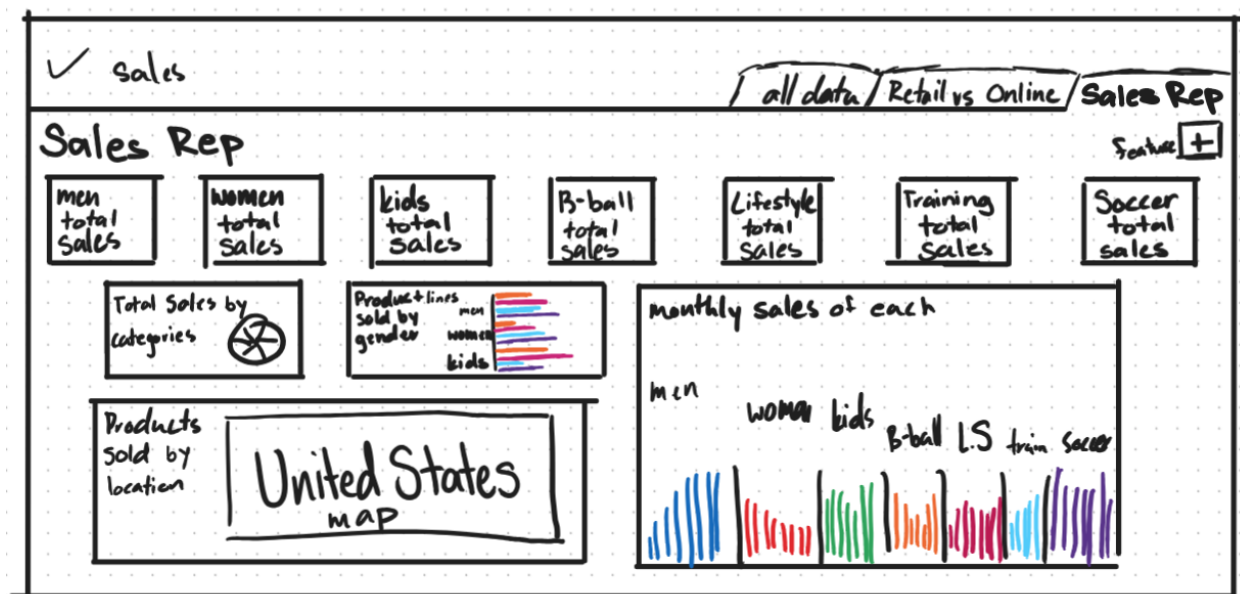


Figure 4. Third Tab “Sales Rep”

5.1 Development Phase

5.1.1 Dataset Implementation

Dataset Implementation Overview:

The Bike Sales dataset is sourced from Kaggle and is the main source of data that the users will be interacting with on the dashboard. The dataset is synthetic, used for educational purposes, and does not include real Nike customer data. This allows users to analyze and view different representations of data with interactive features.

Dataset Implementation Process:

1. Download the Nike Dataset from Kaggle.
2. Clean the Nike Dataset with Power Query in the PBI application. This includes transforming, removing duplicates, correcting data types, and maintaining consistency.
3. Import the cleaned Nike Dataset into Power BI to create visual representations of data.
4. Make the Nike Dataset read-only to users to prevent unnecessary modifications, maintaining integrity.

5.1.2 Dashboard Implementation

Overview for Dashboard Implementation:

The dashboard is an interactive web-based domain developed in the Power BI application, visualizing the Nike Sales Dataset. It allows users to interact and explore different categories, such as product lines, sales representations, online, and retail, and analyze each sales trend for Nike. The website will have three tabs for users to interact with

and explore the different categories of Nike Sales, while maintaining consistency in the GUI and a smooth user experience throughout each tab.

Dashboard Implementation Process:

1. Import the cleaned Nike Sales Dataset into Power BI.
2. Design charts, graphs, KPIs, etc. to display sales trends.
3. Implement slicers and filters as interactive features for users to use to analyze data.
4. Sort the visual representations of the dataset into 3 main tabs: All Data, Online vs. Retail, and Sales Rep
5. Export the dashboard and embed it into the website.

5.1.3 Data Representation

Data Representation Overview:

The data representation of the Nike Sales data is displayed through various representations, such as graphs, charts, and other KPI indicators. It makes the sales data easier to interpret with interactive features as support for user analysis.

Data Representation Process:

1. Create graphs, charts, and other KPI representations with the Power BI application.
2. Sort each visual representation by its respective categories.
3. Format representations to ensure readability and clarity.
4. Test each visual representation with the Nike Sales Dataset to ensure accuracy and interactivity.

5.1.4 Interactive Features

Overview of Interactive Features:

Interactive features allow users to analyze, manipulate, and explore the website about Nike Sales. Features such as filters, slicers, help buttons, drop-down menus, and tabs are implemented to ensure smooth experiences for both returning and first-time users.

Process of Interactive Features:

1. Implement slicers into each tab category for filtering.
2. Implement drop-down menus and check boxes to apply user-selected features.
3. Add a hover feature throughout the website and each tab, guiding new users.
4. Connect the interactions with the dashboard to allow updates to the visualizations without updating the dataset.

5.1.5 Accessibility and Security

Overview of Accessibility and Security:

Users will have quick and simple access to the dashboard through a stable internet connection and web browsers such as Google Chrome, Microsoft Edge, Safari, and Firefox, with no software installation necessary. It allows users to access the dashboard through different devices, such as mobile or PC.

Security for the Nike Sales website is focused on protecting the integrity of the underlying dataset. The dataset in the dashboard will be read-only for casual users, whereas the developers will have access to it to further clean and transform it in the future.

This ensures that the dataset is not being altered by unauthorized users and reduces potential errors in the data.

Process of User's Accessibility in the Dashboard:

1. Users could be able to hover over the KPI charts or graphs to view the description and ensure that users have a smooth experience in usability and learning.
2. The dashboard is exported and embedded into a website, where it is accessible through Google, Microsoft, Firefox, and Safari (for mobile devices).

5.2 Project Setup

5.2.1 Prerequisite

Casual users:

- A secure web browser (Google, Microsoft, Firefox)
- Stable internet
- Device with stable internet (PC or mobile)

Developers:

- GitHub access
- Stable internet
- Power BI desktop (including Power Query)
- Access to Nike Sales Dataset (Kaggle)
- Stable Internet
- Web Development Environment

5.2.2 Use of Dashboard Steps

1. Get on a PC or any mobile device
2. Go to Google Chrome, Microsoft Edge, or Firefox.
3. Go to the search bar and type the link (<https://asparagus-goose-2rll.squarespace.com/>) and press “Enter.”
 - 3.1 If it asks for a password to access the website, put “SP115” and press the arrow button
4. To view different comparisons of categories, click on the tabs located at the top of the screen
5. To apply filters, press the down arrow box to see additional filter options and select which filter you would like, and press the box button next to the desired filter.
6. If you’re new to the dashboard, hover over the dashboard KPI’s to view the description of the data visualization.

6.0 Test Plan

6.1 Overview of Test Plan

The purpose of this section is to:

- Provide details of the test scope
- Record test responsibilities and the goal of the responsibility
- Report other testing necessary to ensure privacy and a smooth experience for users
- Report test strategies and environments used
- List assumptions and dependencies of the PBI Sales
- Providing information on the entry and exit criteria

- Report approvals, test schedules, and training of the project

6.2 Scope of Test Plan

This section contains records of the testing that is performed by the group for the Nike PBI Sales project. It tests the requirements and functions that were listed in the design (SDD) and requirements (SRS) document. The scope of the project that will be documented is:

- Validation of data representation
- Functionality of the interactive features (slicers, sorters, and filters)
- Functionality of the website
- Usability of the website from other devices

6.3 Progression Test Plan

As we are testing the Nike Sales PBI Dashboard, it is important to ensure consistency and accuracy for the behavior of its function and website integrity. As seen in Table 1, the progression test table displays the objectives that the testers are verifying to meet expectations and requirements throughout the testing phase.

Ref	Function	Test Objective	Evaluation Criteria	X-Ref	P
Function to be tested					
P-01	KPI Summary Cards	Verify the core business metrics correctly on the main dashboard	Total Sales displays 5.02M, Retail Sales displays 2.56M, and Online Sales displays 2.56M.	BR-01	P

Ref	Function	Test Objective	Evaluation Criteria	X-Ref	P
P-02	Sales Channel Pie Chart	Confirms the proportional split between Online and Retail channels	The chart accurately reflects a near 50/50 split, with retail at 51.01%, for the selected period.	BR-02	P
P-03	Product Category Bar Chart	Validates the ranking logic for product performances	Clothing is correctly identified as the top-performing category by quantity.	BR-03	P
P-04	Dual Axis Comparison	Test the side-by-side comparison of sales type by product category	Bar charts correctly toggle/compare Online vs. Retail values for product category(Furniture, Electronics, etc.)	UI-04	P

Table 1. Progression Test Objectives

6.4 Regression Test Plan

Regression testing will be performed if the developers update the underlying dataset or when there are new KPI representation implemented into the dashboard website. New representations, such as charts, graphs, or interactive features, would be a part of the regression testing to ensure that the functionality was not impacted by the updates. In Table 2, it goes into detail about what the objectives are and allows the tester to understand what they are testing and verifying for the Nike Sales PBI Dashboard.

Ref	Function	Test Objective	Evaluation Criteria	X-Ref	P
R-01	Regional Slicer	Ensure all visuals update dynamically when a specific region is selected.	Selecting the “West” region updates Total Sales to 280.93k and filters all sub-charts accordingly.	SR-01	P
R-02	Sales Rep Filter	Verify that selecting specific personnel (Alice, Bob, etc.) highlights their specific revenue.	Selecting “Charlie” updates the dashboard to show a Sales Rep Sales KPI of 66.57K.	SR-02	P
R-03	Pagination Controls	Confirm the integrity of navigation between dashboard layers.	Clicking the “Next Page” arrow successfully loads the Monthly and Yearly sales trend view.	UI-01	P

Ref	Function	Test Objective	Evaluation Criteria	X-Ref	P
R-04	Customer Type Filter	Validate the “New” vs “Returning” logic across existing charts.	Slicing by “New” customers correctly reduces the Total Sales to 2.51M without breaking chart links.	SR-03	P

Table 2. Regression Test Objectives

6.5 Test level responsibility

Test Level	External Party	Project Team
Website Testing		P
Dashboard Testing		P
Interactive Features Testing	P	S
After Embedding Dashboard into Website Testing	S	P
User in Different Devices Testing	P	S
Fully Integrated Website Testing		P

Table 3. Responsibility in Each Testing Level

Note:

P=Primary

S=Secondary

6.6 Test Approach

Test Type	Objectives
Functional Testing	Verify that all slicers (Region, Customer Type) and visuals(Sales Rep Performance, Product Category bar charts) correctly filter and display data as intended.
Regression Testing	Ensures that updates to the underlying data model or new measures do not break existing visuals, such as “Sum of Total Sales by Sales_Channel” or “Product Category” bar chart.
User Acceptance	Confirms that end-users can effectively use the “Customer Type” and “Region” slices to find specific insights.
Performance Testing	Evaluate the loading time of the Power BI report and the responsiveness of visuals when switching between high-volume “Sales Channel.”
Data Validation Testing	Ensure the “Sum of Total Sales” and “Product Quantity Sales” displayed on the dashboard match the underlying source data from the Nike databases.

Table 4. Test Type Approaches

6.7 Build Strategy

The build strategy for the project follows the waterfall method. It begins with acquiring a dataset (Nike Sales) from Kaggle and importing it to PBI. In PBI, PowerQuery is used to clean and transform the recently imported dataset. After the cleaning and transformation, the developer can use the cleaned dataset for generating visual representations for the dashboard and implement interactive slicers, sorters, and filters.

The process would follow the same steps to generate representations for the appropriate categories in each tab, allowing users to interact with or view different insights into the data.

Once the pages are completed, they are exported and embedded into the website. Then the testing step comes in to ensure that the pages are behaving similarly and are displaying the correct data. This ensures consistency, reduces any issues, and data validation throughout each page during the development and testing phase.

6.8 Facility, data, and resource provision plan

6.8.1 Test environment

The required test environment for testers is a desktop/laptop or mobile device. With these devices, testers would need a compatible web browser to access the website. It can be Google Chrome, Microsoft Edge, or Firefox.

For developers, they will need the Microsoft Power BI application. They are validating the dataset while creating the dashboard and ensuring accuracy of the visual representation with the cleaned dataset before publishing to the website.

The availability of the test environment is open throughout the development and testing phase. It can be validated multiple times before release.

6.8.2 Access to other applications

Applications that are required for the testing phase:

Developers:

- Microsoft Power BI
- PowerQuery in PBI application
- Compatible web browser (Chrome, Microsoft Edge, Firefox)

Testers:

- Compatible web browser (Chrome, Microsoft Edge, Firefox)

6.8.3 Testing Requirements

To test the Nike Sales PBI Dashboard, you will need:

- Stable Internet Connection
- Any compatible web browser (Google Chrome, Microsoft Edge)
- A device (desktop or mobile)

6.8.4 Data Requirements

The Nike dataset from Kaggle is necessary for developers to begin data cleaning and transformation in the PBI before testing.

Since the dataset is from Kaggle and is static, the developer and tester have to double-check the cleaned data to ensure that the visual representation is accurate with the KPIs for the dataset in the dashboard.

6.8.5 Resources & Skills

Types of resources and skills necessary for the testing phase:

- Experience in using Microsoft PBI for data validation
- Basic knowledge of data analysis concepts
- Basic knowledge for navigation throughout the website
- Experience in PowerQuery for data cleaning and transformation

6.9 Testing Handover Procedure

When it's time to go from one stage to another, the Nike Sales PBI Dashboard undergoes a simple handover process.

Once the developer begins testing within the PBI application, they test for data validation and visual representations after the data cleaning and transformation. Once testing within PBI is confirmed, the dashboard is exported and embedded on the website.

After the dashboard is embedded, the developer would hand the testing process over to the tester to quality check the dashboard through various web browsers on different devices. If issues are found in either of these testing processes, the developer or tester will notify the developer or tester, so that corrective measures can be taken and the testing process can be repeated.

6.10 Testing Metrics

Testing Metrics of Nike Sales PBI Dashboard:

- **Functionality** – It is based on whether the interactive features and the webpage behave as expected

- Data Accuracy – Make sure that the visual representation represents the cleaned and transformed dataset
- Number of Issues Found – Tracks how many issues were encountered while testing
- Usability – Determine how easy or complex the webpage is for users
- Cross-Platform Compatibility – Ensures that the dashboard behaves the same on different devices

The testing metrics will be captured by hand. Testers will go to the website and test these metrics, and it will help the team to find any areas to improve on. This would make sure that the requirements of the dashboard are met.

6.11 Test Environment Diagram

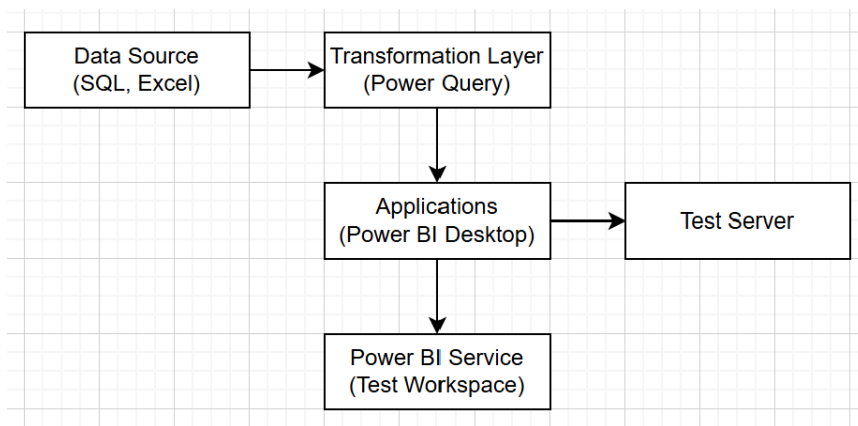


Figure 5. Diagram for Testing Nike Sales PBI Dashboard

6.12 Testers

There will be 2 testers involved in testing:

Tester 1:

- A desktop/laptop
- Stable Internet connection
- Compatible web browser (Google Chrome, Microsoft Edge, Firefox)
- Access to Microsoft Power BI

Tester 2:

- A mobile device (iPhone or Android)
- Stable Internet Connection
- Compatible web browser app (Safari, Google App)

6.13 Hardware and Firmware

Hardware Requirements for testing include a desktop or a laptop, and mobile devices for users to use. The purpose of these devices is to ensure the accessibility and functionality of the website. It is provided by the team members and will solely be used by the team members during the testing phase.

6.14 Software

For the software requirements of the test environment, it will include:

- Access to the Microsoft Power BI Desktop App – For development of the dashboard and dataset validation
- A compatible web browser (Chrome, Microsoft Edge, Firefox) – To access the website and dashboard for accessibility.
- Discord – A chatting application to allow users to send messages if any issues appear while testing

The software should be included in the devices once supplied by the team members.

6.15 Interface

The interface starts once the dataset from Kaggle is downloaded and is imported into the Microsoft PBI application. The dataset could be cleaned and transformed from PowerQuery, and generate visual representations. The interface of the dashboard will be

integrated into the website by the developer. The interface is tested during the testing phase by the developer and the team lead to ensure functionality and proper data flow throughout the website.

6.16 Other Materials

The testing phase tests other requirements that are mentioned in previous documents, such as the Software Requirements Specification (SRS), Software Document Design (SDD), and the Development Document. This ensures that the project team is testing the aligned requirements and design, maintaining consistency from documentation to implementation.

6.17 Environment Control

For software release control, the dataset and the dashboard file will be saved or copied across multiple versions to ensure that all data is not lost. The most recent dashboard will be used to reduce inconsistencies across categories.

For environmental access, it is limited only to the project team members. The users have read-only access to the dataset to prevent unnecessary modifications to the dataset. The team members will have access to the dataset if any updates to the dataset are important and valuable to the users in the PBI application.

For environment monitoring and support, the team will be monitoring and providing support to the dashboard website. In the testing phase, the team ensures that the functionality and usability of the website are at its top performance. If there are issues, the team will notify other members and leave the issue in the thread to ensure that it will be solved by the developer.

6.18 Team Role Responsibilities

Role	Staff Member	Responsibilities
Team Lead, Tester	Mavis Phaphol	Responsible for testing website functionality and users' usability.

Developer, Tester	Mallory Phaphol	Responsible for developing the dashboard and website, and testing to ensure the appropriate functionality of interactive features
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Table 5. Staff's Responsibilities

6.19 Assumptions

Assumptions made while testing the Nike Sales PBI Dashboard:

- The dataset is static and synthetic, so there are no updates made in real-time
- Testers testing the website are assumed to have a stable, good internet connection
- The dashboard created from PBI Desktop is embedded into the website successfully
- Testers are assumed to have basic knowledge of web navigation
- Users are assumed not to have any knowledge of data analysis concepts

6.20 Dependencies

While testing the Nike Sales PBI Dashboard, it depends on certain environments, such as:

- Functionality from the Microsoft PBI for data processing and visualization
- Compatibility of the website on web browsers, such as Google Chrome, Microsoft Edge, Firefox, and Safari, for mobile devices
- Stable internet connection to access and interact with the dashboard
- The development phase for the dashboard should be completed before testing

6.21 Entry and Exit Criteria

Entry	Exit
Acquire Dataset	Import to Power BI Desktop
Cleaned/Transformed Dataset	Creating a Visual Representation
Created a Dashboard of Representation	Representations are in Their Appropriate Location With No Wrong Data
Created Interactive Features	Features are Behaving Correctly, Displaying the Updated Data
Embedding into the Website	The Website is Fully Functioning with No Issues

Table 6. Entry and Exit Criteria Table

7.0 Test Report

7.1 Security

The dataset used for the PBI Sales dashboard is read-only, only allowing the project group with access to apply modifications. The users would not have any access to the dashboard to ensure that no unnecessary updates are applied.

7.2 Stress & Volume Testing (S&V)

S&V testing is not required since the dataset that we are working with is not a large dataset. There are some tests to test how the PBI behaves with the dataset as it is loading on different user devices.

7.3 Connectivity Testing (CT)

The group tested the website dashboard with a stable internet connection to ensure a smooth user experience and was properly loaded on their screen.

7.4 Disaster Recovery/Back Up

There was no disaster recovery or backup testing due to the dataset being static and in development locally.

7.4 Unit Testing

The interactive features for the PBI dashboard were tested individually to ensure that they are behaving correctly and displaying the appropriate representations to users.

The representations of the dataset were tested on itself to ensure that the visualization is not impacted while embedding onto the website.

The different tabs included in the website were tested as well. The test was made to ensure that each tab displayed the correct representation with interactive features to ensure that the appropriate set of data representation is displayed to users.

7.5 Integration Testing

Integration testing was performed when the embedding of the PBI dashboard was created on the desktop application.

7.6 Test Execution Schedule

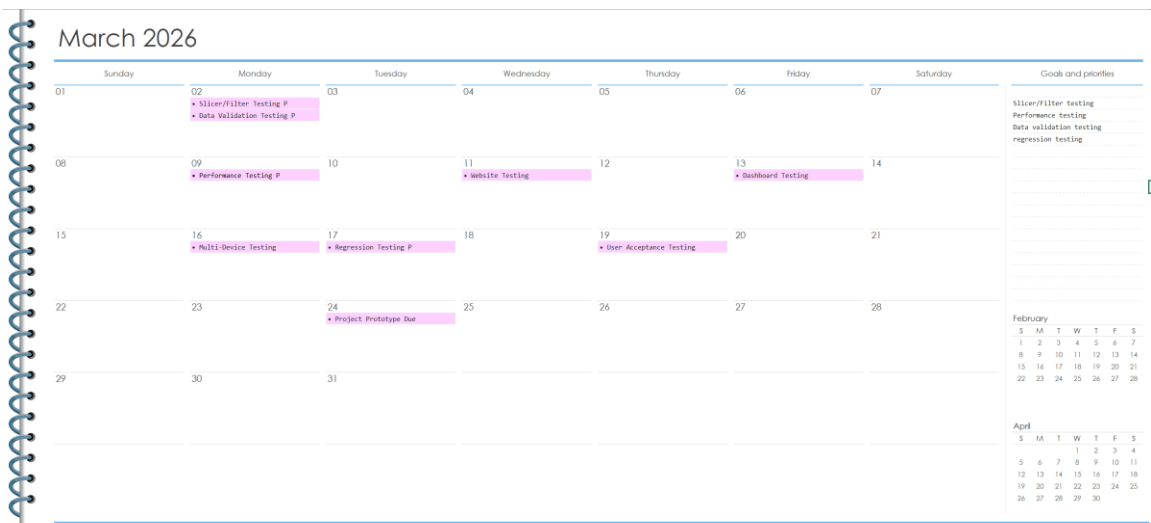


Figure 6. March 2026 Test Schedule

7.7 Testing Tools

Testing tools used for the Nike Sales PBI Dashboard:

Process	Tool
Acquiring Nike Sales Dataset	Kaggle Website
Importing Dataset	Microsoft Power BI
Cleaning/Transformation Dataset	Microsoft Power BI and PowerQuery
Website Functionality	Google Chrome and Microsoft Edge
Integration of Dashboard to Website	Microsoft Power BI and web browsers (Google Chrome and Microsoft Edge)
Website Testing on Devices	Laptop, iPhone, Microsoft Edge, Google Chrome, Safari

Table 7. Tools Used in Each Process

7.8 Establishing Environment

Task	Requirements	Responsibility	Start Date	End Date
------	--------------	----------------	------------	----------

Data Access	Read-only access to the Nike Sales Database	Data Engineer	01/26/26	02/08/26
Workspace Setup	Create “Nike Red – QA” workspace in Power BI Service	Power BI Admin	01/26/26	02/08/26
Website Setup	Generate the report URL and configure the app interface for a clean user testing experience	Dashboard Developer	02/10/26	03/08/26

Table 7. Environments for Each Task Report

7.9 Approvals

The following people are responsible for the critical aspects of testing:

Task	Responsible Person	Escalation/ Approver
Website Integration with Dashboard Completed	Mallory Phaphol	Mavis Phaphol
Fully Integrated Website Testing Completed	Mallory Phaphol	Mavis Phaphol
Interaction Features with Dashboard Visual Behaving Correctly Testing Completed	Mallory Phaphol	Mavis Phaphol

Table 8. Approvals of Task Assigned to the Team Member

7.10 Test Milestones and Schedule

Milestone	Planned End Date	Actual End Date	Person in Charge
Data Validation Completed	02/23/26	02/20/26	Mallory Phaphol
User Acceptance Sign-Off	04/26/26	05/03/26	Mavis Phaphol

Milestone	Planned End Date	Actual End Date	Person in Charge
Production Deployment	04/17/26	05/03/26	Mallory Phaphol & Mavis Phaphol
Visual/UI Testing Completed	03/18/26	04/03/26	Mavis Phaphol

Table 9. Test Milestones Completed by the Team

7.11 Training

Training Requirement	Staff	Date
Slicer and Filter Usage	Mallory Phaphol & Mavis Phaphol	03/30/26
Dashboard Navigation	Mallory Phaphol & Mavis Phaphol	03/30/26
Exporting Data to Excel	Mallory Phaphol	02/02/26

Table 10. Table of Training that was Reported for Staff to Complete

7.12 Defect Management

During our testing phase, if defects are found, we will notify each other of the defect and store it in our group chat. The group chat had a thread function, so we could store all the defects found there. The tool will be the Discord application to store issues found in threads so that the developer can check the thread at any time and resolve the issue.

8.0 Version Control

For each update to the PBI dashboard, the developer would save a copy each time to ensure the security and integrity of the dataset. Since the dataset is synthetic, the dashboard data does not change as often. By saving the dashboard each time, it ensures that the developer is able to revert to the older version in case a problem occurs in the current version. The version control for the dashboard allows us to revert to earlier versions to address any issues the developers want to resolve or to ensure that requirements are being met.

9.0 Conclusion

In conclusion, we discussed multiple topics across the Nike Sales PBI Dashboard. The report began with the design and requirements analysis for the project. The developers and the documentation team developed the requirements and design to provide a smooth experience and easy usability for returning and new users accessing the website. As the team transitioned into the implementation phase, some limitations occurred as we were developing the dashboard and website, such as limited features from using a free PBI account and difficulty in embedding the dashboard into a public space. The limitations did not set the team back long, as we were able to find solutions through different representations for data and publication.

The implementation/development phase was reported as seen in the development section, with processes for each component of the Nike Sales PBI Dashboard and their overview. There are steps and prerequisites that users can follow to ensure a smooth user

experience. Once the development phase had concluded, the team began the testing phase. The test planning section discussed what the testers were testing for and ensured it had met the project's requirements and expectations from the team. The test plan section was followed by the test report, which contains results and approvals from the testers and the team lead. The testing process for each requirement was recorded and documented in the report section. As the project was developing and undergoing multiple tests, it was saved under different file names to ensure that the dashboard was up-to-date and gave developers a safety net in case of corruption or any problems that had occurred in the current version.

The Nike Sales PBI Dashboard was developed to provide users with a learning experience about data analysis through a dataset sourced from Kaggle. Users can interact with the dashboard by using the filters and slicers or hover over the dataset to understand what the representations are for each data point. The dashboard is accessible through a website and a stable internet connection, allowing users to interact and view different infographics about the dataset. This final report provided all the details about the Nike Sales PBI Dashboard's development, test plan, and report as the project is wrapping up its final testing and ensuring the requirements are met and satisfied before releasing it to the public.

Appendix

Glossary

Term	Definition
PBI	Power BI

KPI	Key Performance Indicator
Synthetic	Produced/generated artificially by an algorithm to mimic real-world data
SRS	Software Requirements Specification
SDD	Software Design Documentation
STP	Software Test Plan
STR	Software Test Report
PC	Personal Computer
Kaggle	A website owned by Google, providing users with different types of datasets for Machine Learning, data analysis, and competition participation